

SAI VARDHAN MAKKAPATI

GEN AI ENGINEER

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PROFESSIONAL EXPERIENCE

Main Flow Services and Technologies Pvt. Ltd

Oct 2024 - Jan 2025

- Designed and deployed an AWS-hosted LLM-based virtual interview platform automating candidate screening to final interviews.
- Implemented keyword + semantic resume–job description matching using huggingface embeddings, replacing rigid ATS filters with softer thresholds to improve candidate recall.
- Built a two-stage screening pipeline with lightweight filtering to optimize LLM calls, followed by LLM-powered resume scoring to shortlist top candidates.
- Integrated an autonomous LLM interview agent with structured, rubric-driven evaluations, leveraging AWS EC2, Lambda, API Gateway, and S3.

TECHNICAL SKILLS

- Python, SQL, Bash
- TensorFlow, PyTorch
- LangChain, LangGraph, LlamaIndex, Hugging Face Transformers
- Streamlit, Gradio, FastAPI, Flask
- ChromaDB, Pinecone, Weaviate, Qdrant.
- Open-source and paid LLMs (LLaMA 2/3, Mistral, Falcon, GPT-4/4o,, Google Gemini Pro)
- LoRA/QLoRA fine-tuning, RLHF (PPO, DPO)
- Diffusers, DALL-E, CLIP, BLIP-2, Whisper
- DataStax Cassandra DB (production), MySQL, MongoDB, PostgreSQL
- AWS Bedrock, AWS SageMaker, AWS (EC2, Lambda, API Gateway, S3), Docker, Kubernetes
- Prompt Engineering, vector embeddings, Retrieval-Augmented Generation (RAG), NLP, multimodal AI.

PROJECTS

Retail Sales Data Analysis using SQL

- Designed and managed a relational database for a large retail dataset (800k+ records).
- Wrote complex SQL queries (joins, window functions, CTEs) to extract customer purchase patterns, top-selling products, and seasonal sales trends.
- Created summary tables and views to support reporting and business insights.

YouTube RAG Chatbot with Streamlit UI

- Built a Retrieval-Augmented Generation (RAG) chatbot leveraging LangChain, Hugging Face Transformers, and Groq LLM for multi-turn, context-aware conversations.
- Automated YouTube transcript extraction, chunking, and semantic similarity search using FAISS with vector embeddings (Hugging Face/ChromaDB).
- Developed a fully responsive Streamlit interface supporting real-time chat, session memory, dynamic LLM configuration, and user-friendly interactions.
- Optimized for low-latency retrieval with Dockerized microservices and tested deployment on AWS SageMaker & Bedrock for enterprise readiness.

Multimodal Storytelling Platform using LLMs & AI-Generated Images

- Designed an interactive storytelling platform combining fine-tuned LLaMA-based LLMs for narrative generation and image synthesis models (Stable Diffusion, DALL-E, Diffusers) for visualization.
- Implemented a modular agent-based architecture where AI agents handled plot construction, character design, and scene visualization, coordinated via LangChain/LangGraph workflows.
- Applied LoRA/QLoRA fine-tuning and prompt engineering techniques to improve creativity, narrative flow, and personalization in generated stories.
- Built a simple yet extensible UI with Gradio/Streamlit, enabling users to input prompts and orchestrate multimodal outputs (text + images) seamlessly.

EDUCATION AND CERTIFICATIONS

Bachelor of Technology

M.S Ramaiah University of Applied Sciences, peenya, banglore - 560058
Certificate in Artificial Intelligence and Machine Learning
2021 - 2025

- Introduction to Generative AI – Google Cloud
- Large Language Models & Transformers – Google Cloud
- Generative AI Studio – Simplilearn
- Generative AI for Beginners – Great Learning

- LangChain Basics for Beginners – Great Learning
- Retrieval-Augmented Generation (RAG) – Great Learning
- AI Fundamentals & Prompt Writing – IBM SkillsBuild
- AI Fluency Course – Anthropic
- Career Essentials in Generative AI – Microsoft / LinkedIn